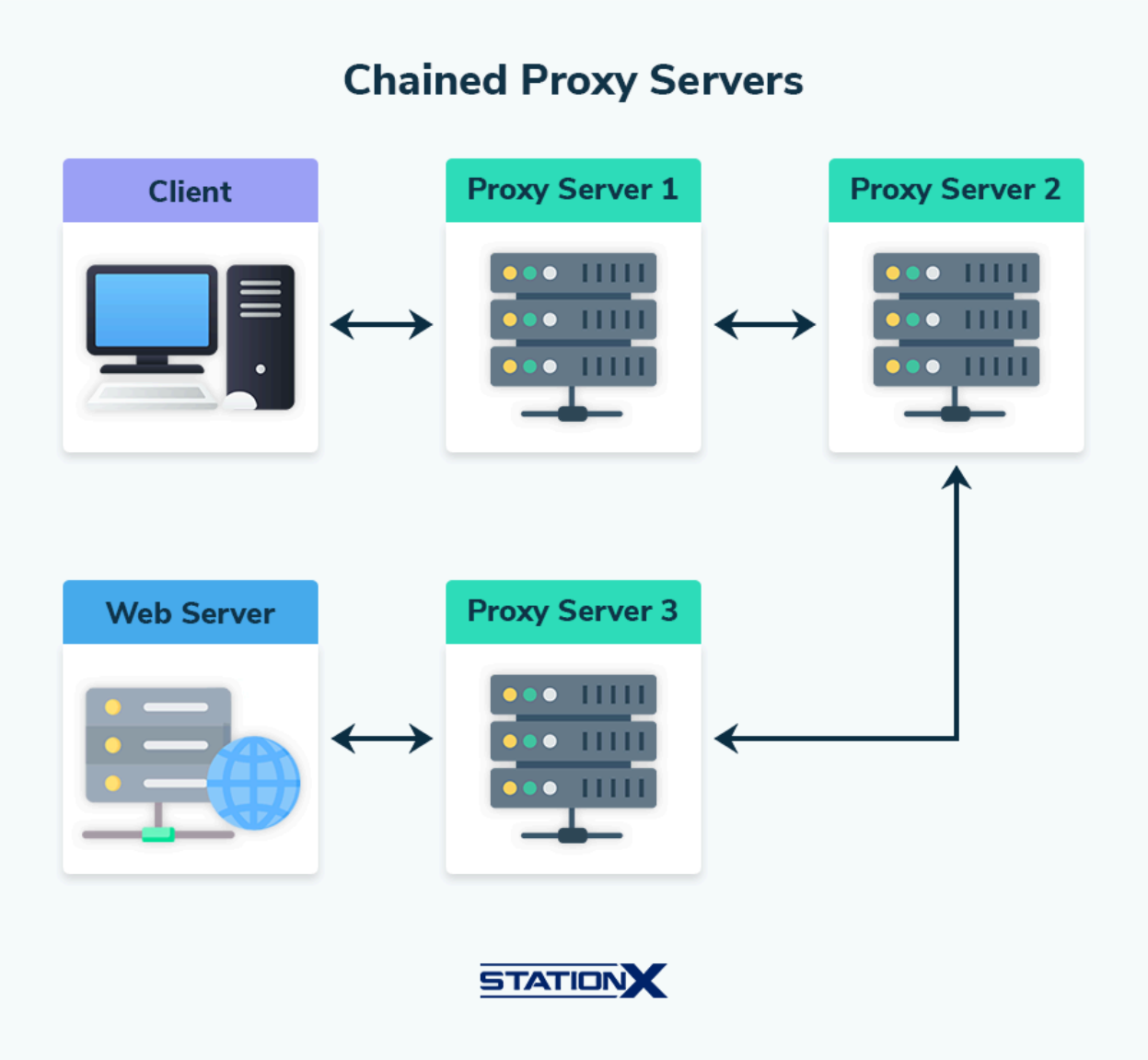
Using Proxychains for Anonymous Hacking



Lab Objective:

Learn how to use proxychains for anonymous hacking by redirecting traffic through proxy servers and Tor.

Lab Purpose:

Proxychains is an open-source tool that allows you to redirect TCP connections through proxies such as SOCKS4, SOCKS5, and Tor. It is useful for hiding the source address of your traffic, evading IDS (Intrusion Detection Systems) and firewalls, and enhancing anonymity during network activities.

Lab Topology:

You will use **Kali Linux** in a **VirtualBox** virtual machine for this lab. Ensure that your VM has internet access (i.e. NAT or Bridged Adapter) and that VirtualBox Guest Additions are installed for better performance.

Phase 1 - Getting Started with Proxychains

Lab Walkthrough:

Task 1: Update Proxychains

1. **Why Update Proxychains?**
    - It is necessary to update Proxychains to ensure you have the latest features, bug fixes, and security patches. Outdated software may have vulnerabilities or compatibility issues.
2. **Steps to Update Proxychains:**
    - Open a terminal in Kali Linux.
    - Switch to the **root user**:

      sudo su
    - Update the package list and install the latest version of Proxychains:

      apt update - (takes a while)
    - Force Kali to upgrade if it will not update:

      apt full-upgrade
    - Install Proxychains using the command:

apt install proxychains4

- Verify the installation by checking the help screen: (i.e.Linux is case sensitive)

  proxychains4 --help

Task 2: Install Tor

1. **Purpose of Using Tor with Proxychains:**
   - Tor enhances anonymity by bouncing your traffic through multiple servers (nodes) around the world, making it difficult to trace the original source of the traffic. When combined with Proxychains, it ensures that all your traffic is routed through the Tor network.
2. **Install Tor:**
   - Run the following command to install Tor:

     apt install tor

Task 3: Configure Proxychains

1. **Modify the Proxychains Configuration File:**
   - Open the configuration file using the nano editor:

     nano /etc/proxychains4.conf
   - Make the following changes:
     1. **Enable dynamic_chain:** Remove the # at the beginning of the line:

        dynamic_chain

        - **Significance:** dynamic_chain allows Proxychains to use a dynamic list of proxies. If one proxy fails, it will try the next one in the chain, providing flexibility and reliability.

     2. **Disable strict_chain:** Add a # at the beginning of the line:

        # **strict_chain**

        - **Significance:** strict_chain requires all proxies in the chain to be available. Disabling it ensures that the tool can still function even if some proxies are unavailable.

3. **Enable proxy_dns:** Remove the # at the beginning of the line:

   proxy_dns

4. **Change Socks4 to Socks5:** At the bottom of the file, change the proxy type to Socks5 and set the IP and port to 127.0.0.1 and 9050 (Tor's default port):

   socks5 127.0.0.1 9050

- Save the file and exit the editor (Ctrl + O, then Ctrl + X).

2. **Verify Configuration:**

   - Run the following command to display the configuration file without comments:

   grep -v "^\\#" /etc/proxychains4.conf

Task 4: Start Tor and Test Anonymity

1. **Start the Tor Service:**

   - Start the Tor service and check its status:

   service tor start

   service tor status

   - Ensure that the service is active and running.

2. **Test Your IP Address:**

   - Open a browser and navigate to https://dnsleaktest.com to check your original IP address.
   - Close the browser and run the following command to open Firefox through Proxychains: (Do not run the following command as **root** - If you are root then type command '**exit**')

   proxychains firefox https://dnsleaktest.com

   - If firefox didn't work try installing a text based browser such as Lynx or w3m: **apt install 'lynx' or apt install 'w3m'**
   - Compare the IP address and location shown on the website. The new IP should be different from your original one and should not be in your home country.

Task 5: Perform Anonymous Network Queries

1. **Run a whois Query:**
    - Use Proxychains to perform a whois lookup anonymously:

      proxychains4 whois example.com

    - Analyze the results. The query should be routed through the Tor network, making it difficult to trace back to your original IP address.

Phase 2 — Privacy Reality Check

1. DNS Leak Investigation (Most Practical Add-On)

You have already visited dnsleaktest.com. Now make it deeper:

Activity:

1. Run the following command in the terminal

   curl ifconfig.me

   What are your results?

2. Then run the command:

   proxychains curl ifconfig.me

   What are your results?

3. Now test the DNS Resolution:

   nslookup google.com

   What are your results?

4. Next run the command:

   proxychain nslookup google.com

What are your results?

Discussion:

- Why does DNS sometimes leak even when IP changes?
- What does `proxy_dns` actually protect?

Tor Identity Rotation Challenge

- Run command in the terminal:

sudo service tor restart

Then run:

proxychains curl ifconfig.me

Discussion:

- **Did your exit IP change?**
- **Why does Tor give you a different "identity"?**

Now, run firefox through proxychains:

proxychains firefox https://coveryourtracks.eff.org

Observe:

- **Screen size**
- **Fonts**
- **Browser version**

- **OS clues**

Task 6: Lab Questions

1. Why is it necessary to update Proxychains before beginning the lab? Explain the steps involved.
2. What is the purpose of using Tor with Proxychains, and how does it enhance anonymity?
3. Describe the process of modifying the Proxychains configuration file. What is the significance of enabling dynamic_chain and disabling strict_chain?
4. Why do you need to change the Socks4 proxy setting to Socks5?
5. After starting the Tor service, use the dnsleaktest.com website. What differences do you observe between your original IP address and the one shown after routing through Proxychains?
6. Run the proxychains whois example.com command and analyze the results. How does Proxychains affect this network query?
7. Discuss potential real-world applications for using Proxychains in cybersecurity. In what scenarios would this tool be most useful?
8. What are the limitations of Proxychains and Tor in evading network security mechanisms like firewalls and IDS systems?
9. Reflect on the ethical implications of using tools like Proxychains and Tor. When would it be appropriate to use these tools, and when could they be misused?